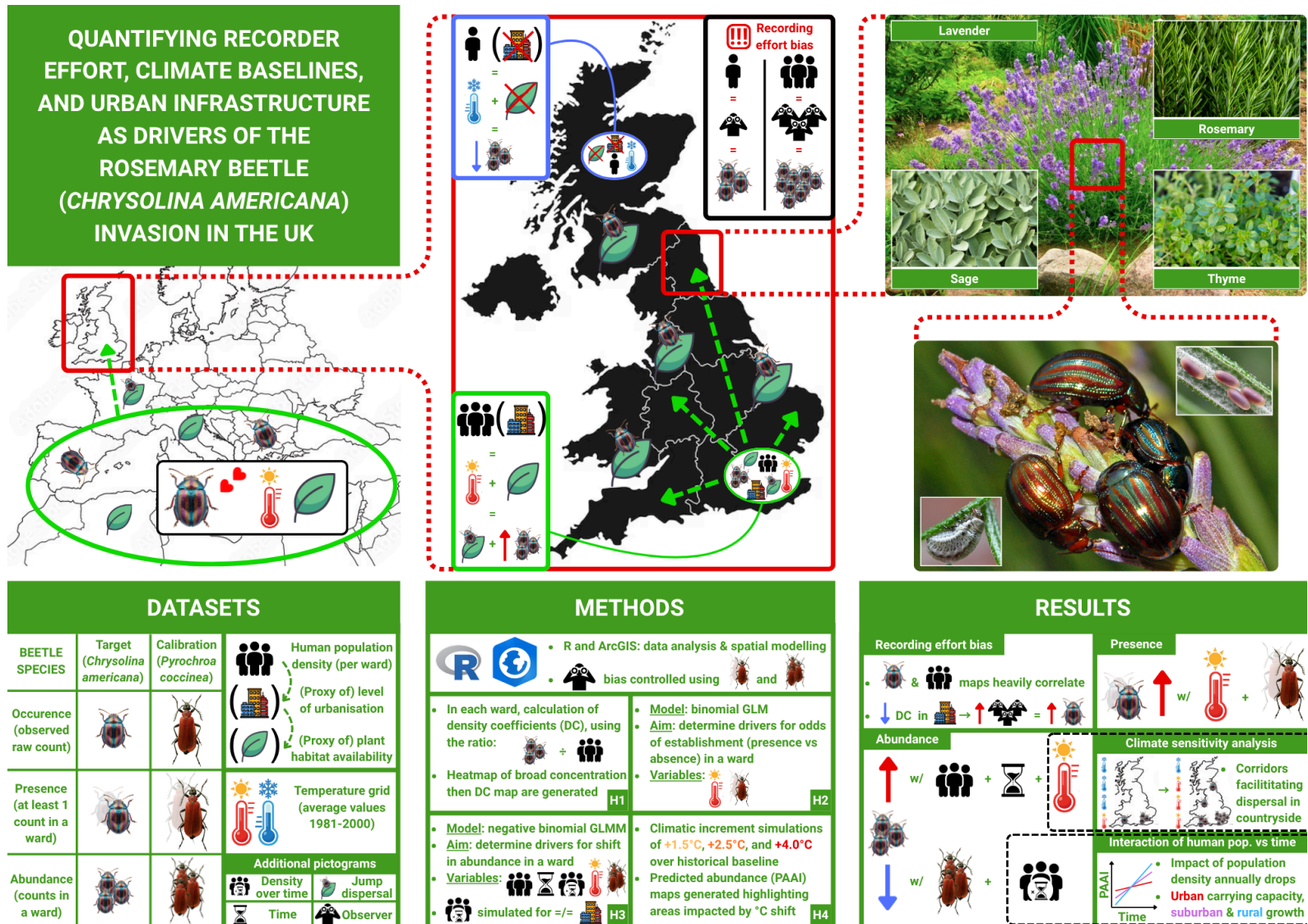


QUANTIFYING RECORDER EFFORT, CLIMATE BASELINES, AND URBAN INFRASTRUCTURE AS DRIVERS OF THE ROSEMARY BEETLE (*CHRYSOLINA AMERICANA*) INVASION IN THE UK.

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A. CONTRIBUTIONS

This research was conducted within the scope of a Master of Science thesis, in partnership between the degree awarding institution of Imperial College London (with Dr A. Vogler as the course director), and the Natural History Museum of London, under the supervision and guidance of Mr M. Barclay (as the senior curator of the Coleoptera collection, and with support from his team and facilities), who made it possible.

We thank the Royal Horticultural Society, the various contributors for the *P. coccinea* records, the Met Office, the Office for National Statistics, the Scottish Government, and the Northern Ireland Statistics & Research Agency for the datasets provided. Furthermore, the R Core Team, Posit, and Esri for the use of their software, all fully cited in the Reference List section.

All datasets and tools mentioned above were procured externally, while the data processing (handling, cleaning, combination, analysis and rendering) was designed for the purposes of this study.

B. AI USE DISCLOSURE

We notify of the use of generative AI (Gemini by Google) for this project, in the following aspects:

- Concepts and planning: outlining the major notions, and adding context prior to the manual review of primary literature.
- Technical and analytical assistance: partially assisting with code and its troubleshooting. The script is human-reviewed, and available in the Supplementary Information section at the end.
- Language and editing: correcting and clarifying grammar, reviewing sentence and paragraph structures to remain cohesive. No content has been generated nor rephrased from the use of AI.

I, the author, confirm that any use of genAI tools is in accordance with Departmental guidelines on genAI use and it has been disclosed. All submitted work has been reviewed and edited by me, has reflected my own understanding, and no content generated by AI has been presented as my own work. I take full responsibility for the academic integrity of the final submission.

C. MISCELLANEOUS

We declare no conflicts of interest, ethical concerns, nor external funding associated with this project.

We strongly recommend consulting the interactive presentation that pairs with this project, created using Esri's ArcGIS StoryMaps. This platform allows for an easier visualisation and exploration of the datasets and results, with a higher degree of resolution thanks to vector-based cartography, while offering a more accessible and condensed approach to the topic. It can be found on the GitHub repository page associated with this project, in the README file.

([link to GitHub](https://github.com/K0rosan/MSc-TBE-Thesis---Rosemary-Beetle), <https://github.com/K0rosan/MSc-TBE-Thesis---Rosemary-Beetle>)

GLOSSARY

- *C. americana* / **rosemary beetle**: *Chrysolina americana*, the target organism.
- **Census area / ward**: Electoral wards in the UK, used as the geographical units of the study.
- **DC** (Density Coefficient): metric addressing the recording effort bias, created to quantify discrepancies between the ratios of observations to recorders in each census area.
- **B-GLM, NB-GLMM** ((Negative) Binomial Generalised Linear (Mixed) Model): statistical models analysing the effect of predictors onto variables, the latter accounting for overdispersion of data.
- **Jump dispersal**: a mechanism of long-distance movement, often human-mediated, followed by the successful establishment of the original dispersing populations.
- **IRR** (Incidence Rate Ratio): statistical metric quantifying the multiplicative change in the expected count of a variable for each one-unit (or one standard deviation) increase in a predictor.
- **Lamiaceae / host plant**: family of aromatic plants (mainly lavender, rosemary, sage, thyme) providing the primary food source and habitat for *C. americana*.
- ***P. coccinea* / black-headed cardinal beetle**: *Pyrochroa coccinea*, the calibration organism used to statistically mitigate the recording effort bias.
- **OR** (Odds Ratio): statistical metric quantifying the multiplicative change in the baseline odds of a binary event occurring, for each one-unit (or one standard deviation) increase of a predictor.
- **PAAI** (Predicted Annual Abundance Index): simulated average number of sightings per year.
- **REB** (Recording Effort Bias): systematic error in datasets where species occurrence records disproportionately reflect human distribution rather than biological reality.
- **SD** (Standard Deviation): measure of the amount of variation or dispersion in a set of values, used in this study to normalise climatic and demographic predictors
- **UHI** (Urban Heat Island): phenomenon where human infrastructure generates, absorbs and retains heat, creating localised microclimates that are warmer than the surrounding areas.

ABSTRACT

Chrysolina americana (Linnaeus, 1758), or the rosemary beetle, is a Mediterranean leaf beetle that has rapidly expanded upon reaching the UK in the 1990s. It is associated with popular garden plants (from the family Lamiaceae) imported through the horticultural trade. The conditions provided by cities, both in temperature and habitat availability, create suitable grounds for its proliferation. Citizen science can be used to census its geographical extent and local abundance, but the impact of the recording effort bias on presence-only datasets skews the data by overestimating abundance near densely populated areas. Statistical tools dissociating biological occurrence versus perceived abundance, such as calibration species, were employed in the generalised linear (mixed) models. The analyses successfully quantified: firstly, the impact of temperature on the odds of establishment of *C. americana*; secondly, its effect on abundance, adding a human population density variable, followed by a temporal interaction analysis. We found most metrics to be positively correlated, except for the calibration species's abundance and the interaction term, the latter revealing an impact of human population density that diminishes over time, driving the acceleration of growth rate away from the urban hubs due to approaching carrying capacity. Predicted abundance was then spatially rendered to represent alternate climatic scenarios, showing the formation of ecological corridors associated with warming. The communication of invasion ecology, through the case of *C. americana*, can root the understanding of environmental predictors based on public-driven datasets, increasingly becoming valuable components of habitat/urban management programs, requiring more thorough research.

Keywords: *Chrysolina americana*, invasion dynamics, climate change, urban heat island, recorder bias.

I. INTRODUCTION

A. BACKGROUND

A. 1. Invasion ecology and the rosemary beetle

The successful colonisation of territories by invasive species presents a major challenge in the preservation of habitats and their biodiversity. Shifts in resource availability may limit access by established, native organisms, leading to ecosystem degradation, and potential economic challenges (Acheampong and Opoku, 2023). Geographic and climatic constraints often act as the primary barriers in restricting the expansion of populations, defining an organism's range (Stock, 2005). Rapid climatic change, especially caused by human activity, alters such limits so they overlap, thus turning previously unsuitable habitats for certain species to settle in, and exploit the resources found in that new range, often at the cost of native ones (Marangi *et al.*, 2026). The monitoring of expanding populations becomes increasingly important in the context of ecosystem management, in order to predict growth trends more accurately in a changing climate.

Chrysolina americana (Linnaeus, 1758), also known as the rosemary beetle, is a member of the leaf beetles (family Chrysomelidae, order Coleoptera). While chrysomelids are phytophagous, this species feeds on specific plants from the family Lamiaceae with high camphor content (Yang *et al.*, 2023), namely lavender (*Lavandula spp.*), thyme (*Thymus spp.*), sage and rosemary (*Salvia spp.*), which are popular garden plants. *C. americana* is presently considered a non-native invasive species, originating from the Mediterranean region (Hadjiconstantis and Zoumides, 2021), spreading to the UK in the 1990s through the ornamental plant trade (Smith *et al.*, 2007), and now established in the domestic gardens across the country (Bieńkowski and Orlova-Bienkowskaja, 2018).

A. 2. Cities as drivers of dispersion and thermal havens

C. americana, while its flight capabilities are discussed (Beenen and Roques, 2010; Friedman, 2016), expands across the UK mainly through jump dispersal, defined as the long-distance movement of individual organisms, followed by the successful establishment of the original dispersers (Pielou, 1979). This “city-hopping” is often human-mediated (Suarez *et al.*, 2001), driven by the commerce of their host plants, which may carry eggs located underneath their leaves, or larvae and adults commuting between the branches and the surrounding soil as part of their life cycle (Datiles and Acevedo-Rodríguez, 2014).

The urban environment is thermally a suitable habitat for the rosemary beetle: cities emit, reflect, and capture heat. Dark asphalt roads and tall concrete buildings are often sinks that block air flow, and trap heat coming from cars, houses, and sunlight (Phelan *et al.*, 2015). This phenomenon, known as the urban heat island (UHI) effect, increases the average temperature of cities, creating a climate more in line with the original range of *C. americana*, and improving survival rate across colder seasons (Cox, 1994). The refuge of cities favours settlement on the territory, primarily through warmth and food availability.

A. 3. Citizen science and its role in population ecology

Traditional expert-led biodiversity monitoring is often limited by the quantity of information that can be retrieved and processed by a select few individuals (Agnello *et al.*, 2022). Such operations can be outsourced to larger, public-driven online platforms such as iNaturalist, GBIF, or the NBN Atlas, especially for census datasets. While the alternative of citizen science solves the issue of sampling size, it introduces drawbacks associated with this democratisation.

Some issues can be addressed between the acquisition of raw data points and their integration to a refined repository, such as consistency across recorders of heterogeneous methodologies and skill level (reliability of identification), or the quality of data (completeness of details like date/location), which can be reviewed by reputable organisations before integrating datasets (Wiersma *et al.*, 2024). Meanwhile, some are inherent to observation data as a collection method: it is by nature presence-only, meaning such information tells where a species does occur, but cannot tell where it does not (Hermoso *et al.*, 2015). Another complication is the clustering of data points where recorders are more likely to be present, translating into more opportunities of observations in and around accessible areas like cities (Lipiński, 2026). This is known as the recording effort bias (REB), which influences the interpretation of datasets if not accounted for, both as a first impression, and in statistical studies.

B. STUDY RELEVANCE

The rosemary beetle, while usually not lethal to healthy plants (Friedman, 2016; Hadjiconstantis and Zoumides, 2021) and thus not representative of an ecological urgency, is a model organism of choice when studying invasion dynamics, especially for a species that performs jump dispersal. Its dissemination across several distant UK cities offer independent opportunities for the beetle to thrive from an epicenter, allowing localised, non-overlapping studies of its expansion and population growth rate.

Monitoring *C. americana*'s expansion in the UK fills a knowledge gap in the study of leaf beetles and their dispersal, which is currently understudied (Bieńkowski and Orlova-Bienkowskaja, 2018). Elucidating its population dynamics and their environmental influences can provide valuable insights into the broader mechanisms of ecological invasions. The extrapolation of observed patterns and methods to other organisms is useful for ecologically or behaviourally similar species, and can still shed a light on the effect of sampling biases, or (a)biotic interactions for broader taxa (Yates *et al.*, 2018).

C. AIMS & OBJECTIVES

The study aims to quantify the environmental and anthropogenic drivers responsible for the rosemary beetle invasion, by measuring in isolation the different variables and their effect on the beetle's presence and abundance, before visually representing the effect of its main predictors' modulation.

To proceed, the objectives will be associated to their respective hypothesis (**H**):

- To map the raw current distribution of the rosemary beetle, while addressing and measuring biases associated with the citizen science recording process (**H1**).
- To mathematically isolate the effect of climate from the effect of the recording effort bias, using a calibration species (*P. coccinea*) as control, to determine the initial thermal establishment of *C. americana* (**H2**).
- To quantify the density-dependent expansion of the population by assessing how localised human density amplifies *C. americana*'s abundance and dictates its carrying capacity (**H3**).
- To simulate and visualise the current population distribution through an alternate scenario of uniform climate warming, starting from a historic baseline, and increasing temperature incrementally to the +1.5°C, +2.5°C, and +4.0°C thresholds (**H4**).

D. QUESTION & HYPOTHESES

How do observer effort, historical thermal baselines, and urban density drive the *C. americana* invasion, and how does its current distribution respond to simulated uniform warming?

- **H1**: To answer, we firstly hypothesise that the perceived abundance of *C. americana* is severely localised due to the recording effort bias, artificially concentrating observations in cities, and requiring calibration via a control species (*P. coccinea*).
- **H2**: Secondly, the initial establishment of the rosemary beetle is constrained by a minimum regional climatic temperature threshold.
- **H3**: Thirdly, their concentration in urban hubs increases locally due to both temperature and human population levels, in which it approaches carrying capacity before dispersing outwards in the suburban and rural environments.
- **H4**: Finally, temperature increase simulations will amplify the abundance of the beetle in cities as they warm up, while enabling previously uncolonised zones to be habitable, creating corridors for its spread.

II. MATERIALS & METHODS

A. DATASETS & SOFTWARE

A. 1. Observations of *C. americana*

The first biological dataset (fig. 1a) is the distribution of the rosemary beetle (*Chrysolina americana*) over the UK between 1994 and 2021, retrieved from the Royal Horticultural Society's monitoring scheme (Royal Horticultural Society, 2022), collecting 29,099 occurrences (28,062 usable ones after filtering out rows lacking year, or coordinates).

A. 2. Observations of *P. coccinea*

The second biological dataset (fig. 1b) is the distribution of the black-headed cardinal beetle (*Pyrochroa coccinea* (Linnaeus, 1761)) in the UK (GBIF.org, 2026). It contains 3,752 usable observations sourced from various recording schemes, still updated to this day.

P. coccinea is a native species of fire-coloured beetle (family Pyrochroidae), whose bright red colour makes it conspicuous, more likely to be recorded. Its distinct habitat (woodlands) from the target species helps determine which areas are actively assessed by recorders for both: areas with records of *P. coccinea* but lack of *C. americana* show that an area is surveyed, but the latter is genuinely absent. Lack of both signify either the absence of recorders in an area, or true biological absence. Thus, *P. coccinea*'s role as a covariate dissociates the predictors from the REB, by quantifying change in detection/fluctuation of *C. americana* populations unrelated to temperature, population density, or time.

A. 3. Temperature grid

The environmental dataset (fig. 1c) is the Met Office's Annual Average Temperature Change and Projections (Met Office, 2025). Each grid square (12×12 km) contains the value of the average temperature recorded between the years 1981 and 2000.

A. 4. Census areas of the UK

Finally, the geographic and demographic dataset (fig. 1d) is an amalgamation and harmonisation of geospatial and census datasets, due to the lack of a unified macrodata for the UK: the 2022 population estimates, aligned with the 2023 ward boundaries, retrieved from the Office for National Statistics (ONS, 2025) for England and Wales; the 2022 Intermediate Zones (Scottish Government, 2026); finally, the 2021 Super Data Zones, from the Northern Ireland Statistics and Research Agency (NISRA, 2023).

A. 5. Software

R version 4.5.2 (R Core Team, 2026) was run in the Positron IDE (Posit Team, 2026), using the packages *dplyr* for data manipulation, and *sf* for geometric calculations. ArcGIS Pro 3.7, ArcGIS Online, and ArcGIS StoryMaps (Esri Inc., 2026) were the platforms strictly used for cartographic rendering of the raw and processed spatial datasets.

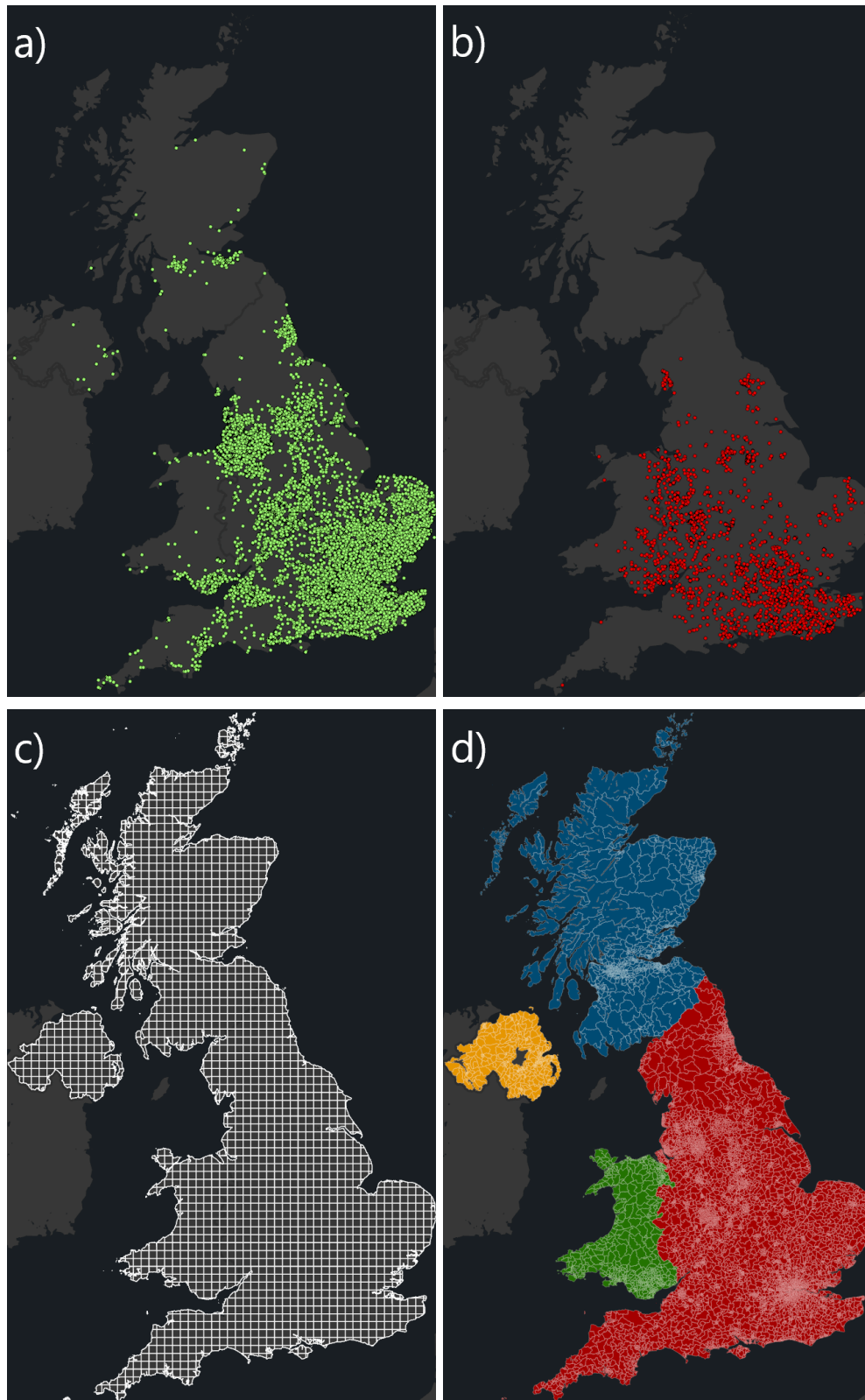


Figure 1: Cartographic representations of the study's raw datasets: **(a)** RHS's observations of *C. americana*, **(b)** observations of *P. coccinea* (various sources), **(c)** Met Office's temperature grid, and **(d)** Wards, Intermediate Zones, and Super Data Zones of the UK, representing the study's census areas.

B. OBSERVER EFFORT & DENSITY MAPPING (H1)

B. 1. Mapping raw observations over time

To show the influence of the REB in cities, we mapped the rosemary beetle's cumulative observations over time, showcasing the most frequent areas of encounter. The data points were individually mapped using their coordinates, before being translated into a point density surface (a heatmap) to express concentration over a general area. Resulting maps are visualised on a 3-year basis starting from 2003.

B. 2. Calculating the density coefficients of census areas

We mathematically demonstrated the effect of the REB by evaluating the proportion of observations to potential recorders in a ward, and comparing it to other wards. Each census area's population (*Total_Pop*) was divided by its surface area (*Area_SqKm*) to output the human population density per km² (*Pop_Density_SqKm*), which was then evaluated against its observations (*Total_Ca*) to create a ratio (*Density_Coefficient*) quantifying beetle density relative to urban presence:

- $Pop_Density_SqKm = Total_Pop / Area_SqKm$
- $Density_Coefficient = Total_Ca / Pop_Density_SqKm$

Maps of the three last variables were generated in ArcGIS Pro using their corresponding columns. Their continuous scale was divided into 10 classes, using the Jenks natural breaks method, excluding true zeros to reflect the absence of *C. americana*, and consequently of a density coefficient (DC).

C. MODELING THERMAL ESTABLISHMENT (H2)

To determine the probability of initial establishment of *C. americana* in each ward, we evaluated its binary presence/absence (at least one record in any year, or none) against temperature. A binomial generalised linear model, executed using *base::glm()*, was used to determine the relationship between a binary response variable (*Presence_Ca*) and its independent variables:

- *Temp_Hist_Z* is the predictor, the baseline temperature normalised into a Z-score such that a value of 0 corresponds to the UK average.
- *Presence_Pc* is the covariate, the binary presence of *P. coccinea*.

The model is thus defined as follows, where $P(Y = 1)$ is the probability of the the successful establishment of *C. americana* in a ward, and β the fixed-effect coefficients (with β_0 the intercept):

- $\text{logit}(P(Y = 1)) = \beta_0 + \beta_1(Temp_Hist_Z) + \beta_2(Presence_Pc)$

The odds ratios (OR) were then calculated to assess the predictors' effect on the odds of presence of *C. americana* (O), which are then converted to probabilities (P), using baseline odds (BO), odds ratio for temperature (OR_1) and *P. coccinea* (OR_2), and values for *Temp_Hist_Z* value (X_1) and *Presence_Pc* (X_2):

- $O = BO \times (OR_1^{X_1}) \times (OR_2^{X_2})$
- $P = O / (1 + O)$

D. QUANTIFYING URBAN CARRYING CAPACITY (H3)

D. 1. Modeling drivers of abundance

To isolate the factors driving the abundance of *C. americana*, a negative binomial generalised linear mixed model (NB-GLMM) was run under the package *glmmTMB* and its namesake function, *glmmTMB()*. The model handles overdispersion (statistical variance exceeding the mean of a dataset), or in ecological terms the clustering of occurrences in a localised area, as individuals prefer to aggregate around most suitable conditions rather than spread evenly over territories.

To resolve pseudoreplication, the model introduces a random effect (the ward unique ID, *Zone_Code*) to treat the same ward seen across the years as recurring. Recognising 149,100 ward-year combinations (5,325 wards × 28 years) as not independent effectively determines an amount of variance that is unexplained by our study, due to inherent conditions (land-use, landscaping practices, microclimate...) that vary between wards.

The response variable is the ward abundance of *C. americana* (*Count_Ca*). Independent variables are:

- *Year_Centered* is the time elapsed (1994 set as year 0), expressing the yearly increase.
- *Pop_Density_Z* is the proxy for habitat suitability, it is the normalised human population density, such that 0 corresponds to the national average.
- *Temp_Hist_Z*, like in the presence model, is the normalised temperature across the UK.
- *Count_Pc* is the abundance of *P. coccinea* in each ward. This covariate reflects citizen science activity, calibrating for the REB.
- *Year_Centered × Pop_Density_Z* is the interaction term of the study, which measures the effect of human population density over the years on the abundance of *C. americana*.

The model is thus defined as follows, where $E(Y)$ is the expected count of *C. americana*, β the fixed-effect coefficients (with β_0 the intercept), and u_i representing the random intercept for each ward:

- $\ln(E(Y)) = \beta_0 + \beta_1(\text{Year_Centered}) + \beta_2(\text{Pop_Density_Z}) + \beta_3(\text{Temp_Hist_Z}) + \beta_4(\text{Count_Pc}) + \beta_5(\text{Year_Centered} \times \text{Pop_Density_Z}) + u_i$

Each predictor's incidence rate ratios (*IRR*) was then calculated, which can be raised to the power of its respective variable value (X) to assess the impact (N_{new}) of each variable onto a count population (N_{old}):

- $N_{new} = N_{old} \times (IRR_1^{X_1}) \times (IRR_2^{X_2}) \times \dots$

D. 2. Density dependence across the years

Density dependence of *C. americana* is examined under the interaction term, by simulating three wards, each with a distinct *Pop_Density_Z* score (-1 = rural, 0 = suburban, 2 = urban), plotted against time (*Year_Centered*). We isolated the effect of the interaction term by holding the other predictors at 0 (simulated wards have an average temperature, and a baseline level of record for *P. coccinea*).

The function *ggpredict()* (package *ggeffects*) was run to extract the predictors of interest from the NB-GLMM, projecting the predicted annual abundance index (PAAI) using the interaction term value over time. The graph was then rendered using the package *ggplot2*, displayed on a logarithmic scale to obtain slopes representing the PAAI rate of increase, for each scenario.

E. ALTERNATE CLIMATE SCENARIO SIMULATIONS (H4)

To simulate *C. americana*'s abundance if it were to colonise the UK under different climatic conditions, we isolated the effect of temperature from the beetle's chronological expansion, using a temporal lock (set to 2021). This neutralises the invasion's spread due to the passage of time, freezing the population to its maximum observed extent. The REB was also controlled by nullifying *P. coccinea* counts. The baseline prediction was performed onto the NB-GLMM previously generated using `base::predict()`, effectively creating a foundational +0°C increase, on which the scenarios are built upon.

The climate warming scenarios were introduced using `dplyr::mutate()`, by modulating the baseline temperature of individual wards by +1.5°C, +2.5°C, and +4.0°C increments (*Change*), and applying them uniformly to the baseline (*Temp_Hist*). In the same process, we also standardised the simulated temperatures to their own Z-scores, preserving the original historical mean (μ) and standard deviation (σ) to properly treat the alternate climate scenarios as anomalous. The formula is:

- $$Temp_Hist_Z_{new} = ((Temp_Hist + Change) - \mu_{historical}) / \sigma_{historical}$$

Temperature maps were generated in ArcGIS Pro using the predicted abundance for each scenario, including the baseline. The continuous scale was divided into 10 manually defined classes, on a fixed absolute scale to resolve range conflicts between the scenarios.

III. RESULTS

A. RAW OBSERVATIONS & DENSITY COEFFICIENT (H1)

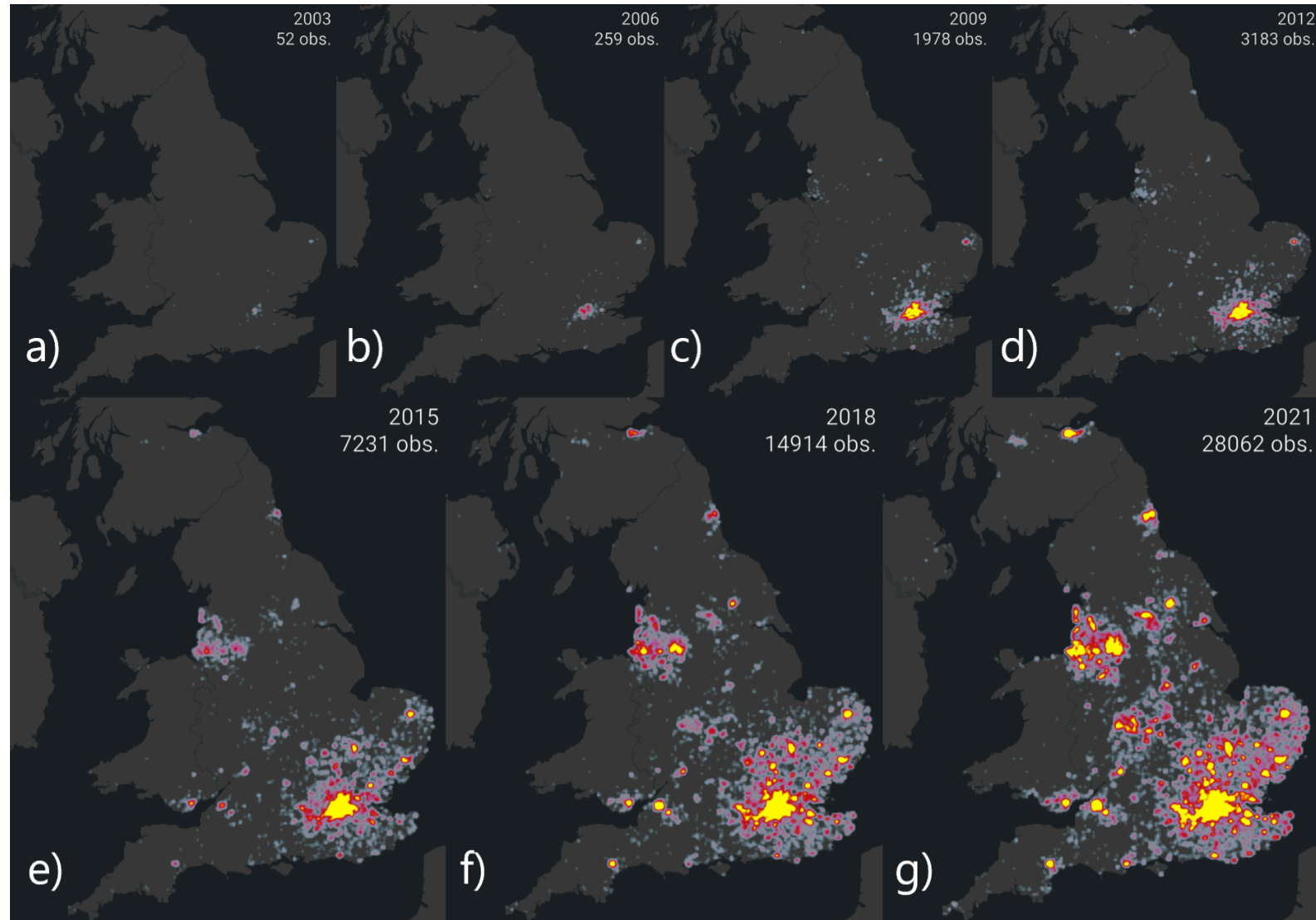


Figure 2: Cumulative observations of *C. americana* in the UK between 1994 and 2021, representing the number of times an encounter with the rosemary beetle has been recorded. **(a)** Occurrences between 1994 and 2003, with each subsequent map **(b-g)** adding 3 years worth of observations.

Initial records of *C. americana* were scarce, and heavily localised in the Greater London area (fig. 2a-c). Over time, the recording rate compounded exponentially, establishing secondary hubs in major northern and coastal cities (Liverpool/Manchester, Bristol, Norwich, Newcastle, Edinburgh), while spatial gaps between colonised areas began to narrow (fig. 2d-g).

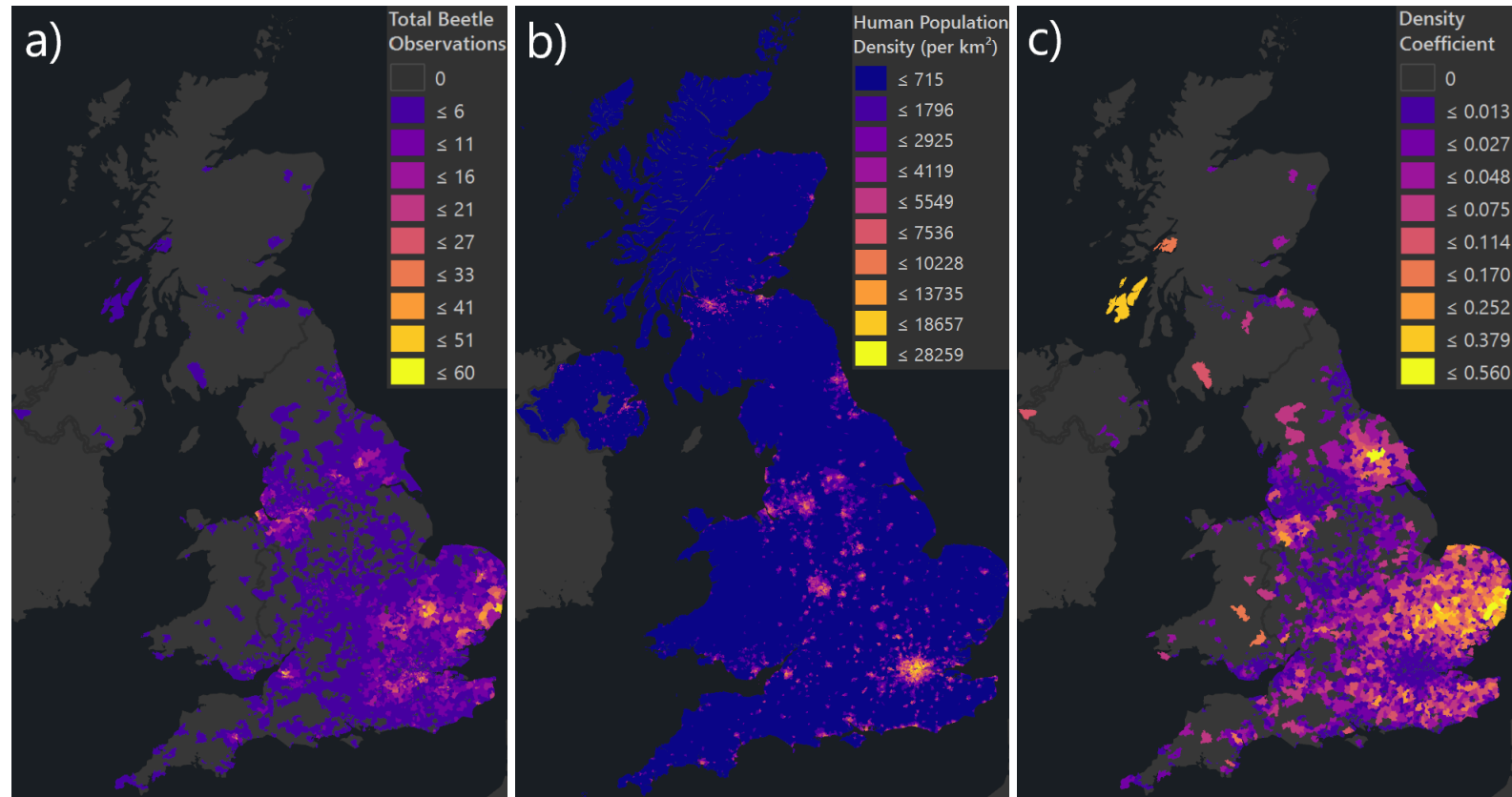


Figure 3: Spatial representation of *C. americana* observations relative to human population metrics in the UK. All values are per ward:

- (a)** Total *C. americana* observations (mean = 2.863, median = 1 observation(s)),
- (b)** Human population density per square kilometer (mean = 2,627, median = 1,871 people/km²),
- (c)** Density coefficient, which is the ratio of the previous metrics (mean = 0.007, median = 0.0002).

Wards with the most observations are generally located in the South East (fig. 3a). UK observations average 2.863 per ward, and concentrate in England, with rare records elsewhere. Unlike *C. americana* population, ward size correlates with human population density, showing populations clustering in the main urban hubs (fig. 3b), London having the smallest but densest census areas.

Fig. 3c emphasises areas where *C. americana* records and human density are disproportionate relative to one another. Their ratio is higher in regions like East Anglia, comprising many wards with a high DC. Wards located West and South of London fluctuate in the middle range of the scale, a pattern that is also visible in Scotland through its scattered observations. At the bottom of the scale, urban centers, especially London, cluster the most (non-zero) lowest-scoring wards of the country.

B. PRESENCE MODEL: THE INITIAL THERMAL THRESHOLD (H2)

Table 1: Summary of the analysis on the impact of the normalised temperature index (*Temp_Hist_Z*), and the co-occurrence of *P. coccinea* (*Presence_Pc*), onto the response variable (*Presence_Ca*), which is the probability of a first establishment of *C. americana* in a ward, using a binomial generalised linear model (B-GLM).

Table 1: Logistic Hurdle Model (Presence Probability)			
Presence Ca			
<i>Predictors</i>	<i>Odds Ratios</i>	<i>CI</i>	<i>p</i>
(Intercept)	0.109	0.107 – 0.112	< 0.001
Temp Hist Z	1.487	1.455 – 1.520	< 0.001
Presence Pc	2.005	1.716 – 2.332	< 0.001
Observations	149100		
R ² Tjur	0.009		

149,100 combinations of ward-year were evaluated to quantify the independent variables driving a first detection of *C. americana* (table 1), all found to be highly significant ($p < 0.001$). With baseline conditions (average temperature and absence of *P. coccinea* observations), the odds of presence were calculated at 0.109 in any given ward (baseline intercept OR = 0.109), corresponding to a baseline probability of 9.8% ($P = 0.098$).

The odds of detection increase by 48.7% (*Temp_Hist_Z* OR = 1.487) for every standard deviation ($SD = 0.859^{\circ}\text{C}$) of temperature increase (and no detection of *P. coccinea*), corresponding to a probability of 13.9% ($P = 0.139$). Furthermore, if *P. coccinea* was detected in the same ward-year combination (with no temperature shift), the odds of detecting *C. americana* double (*Presence_Pc* OR = 2.005), a probability of 18.0% ($P = 0.180$). The probability of establishment of *C. americana* when both temperature increases by one SD and a *P. coccinea* is detected is 24.5% ($P = 0.245$).

C. ABUNDANCE MODEL: A DENSITY-DEPENDENT POPULATION (H3)

Table 2: Summary of the analysis on the multi-variable impact on the abundance of *C. americana* (*Count_Ca*) in occurring wards. The negative binomial generalised linear mixed model (NB-GLMM), coupled with random intercepts (*Zone_Code*), tests for the impact of the following independent variables on the response variable: time passing (*Year_Centered*), normalised human population density index (*Pop_Density_Z*), normalised temperature index (*Temp_Hist_Z*), abundance of *P. coccinea* (*Count_Pc*), and the annual impact of population density (*Year_Centered* × *Pop_Density_Z*).

Table 2: Negative Binomial Count Model (Abundance Index)			
<i>Predictors</i>	Count Ca		
	<i>Incidence Rate Ratios</i>	<i>CI</i>	<i>p</i>
(Intercept)	0.001	0.001 – 0.001	<0.001
Year Centered	1.283	1.278 – 1.288	<0.001
Pop Density Z	1.761	1.671 – 1.855	<0.001
Temp Hist Z	1.618	1.554 – 1.684	<0.001
Count Pc	0.829	0.766 – 0.898	<0.001
Year Centered × Pop Density Z	0.973	0.970 – 0.975	<0.001
Random Effects			
σ^2	1.94		
τ_{00} Zone_Code	0.59		
ICC	0.23		
N _{Zone_Code}	5325		
Observations	149100		
Marginal R ² / Conditional R ²	0.628 / 0.715		

All the modelled drivers of the abundance of *C. americana* (table 2) were found to be highly significant ($p < 0.001$), and setting them to 0 determined a baseline where no rosemary beetles are to be observed (intercept *IRR* = 0.001). The abundance index increases by 28.3% annually (*Year_Centered IRR* = 1.283). The human population density index is positively associated too, each *SD* (*SD* = 3,074 people/km²) increasing populations by 76.1% (*Pop_Density_Z IRR* = 1.761), likewise for temperature at

61.8% ($Temp_Hist_Z$ IRR = 1.618) for every SD ($SD = 0.859^{\circ}C$). However, for every additional record of *P. coccinea*, the abundance of *C. americana* decreases by 17.1% ($Count_Pc$ IRR = 0.829). Finally, $Year_Centered$ and $Pop_Density_Z$ interact negatively, meaning that population density sees its annual impact on beetle population decrease by 2.7% ($Year_Centered \times Pop_Density_Z$ IRR = 0.973), compounding over time.

The effects of overdispersion were accounted for using the baseline variance between census areas ($\sigma^2 = 1.94$) and a residual variance within census areas ($\tau_{00} = 0.59$). The intraclass correlation coefficient ($ICC = 0.23$) shows that 23% of the remaining unexplained variance is driven by inherent differences between zones. Our predictors explain 62.8% of the total variance in beetle abundance (marginal variance $R^2 = 0.628$), rising to 71.5% if factoring for individual census areas (conditional variance $R^2 = 0.715$).

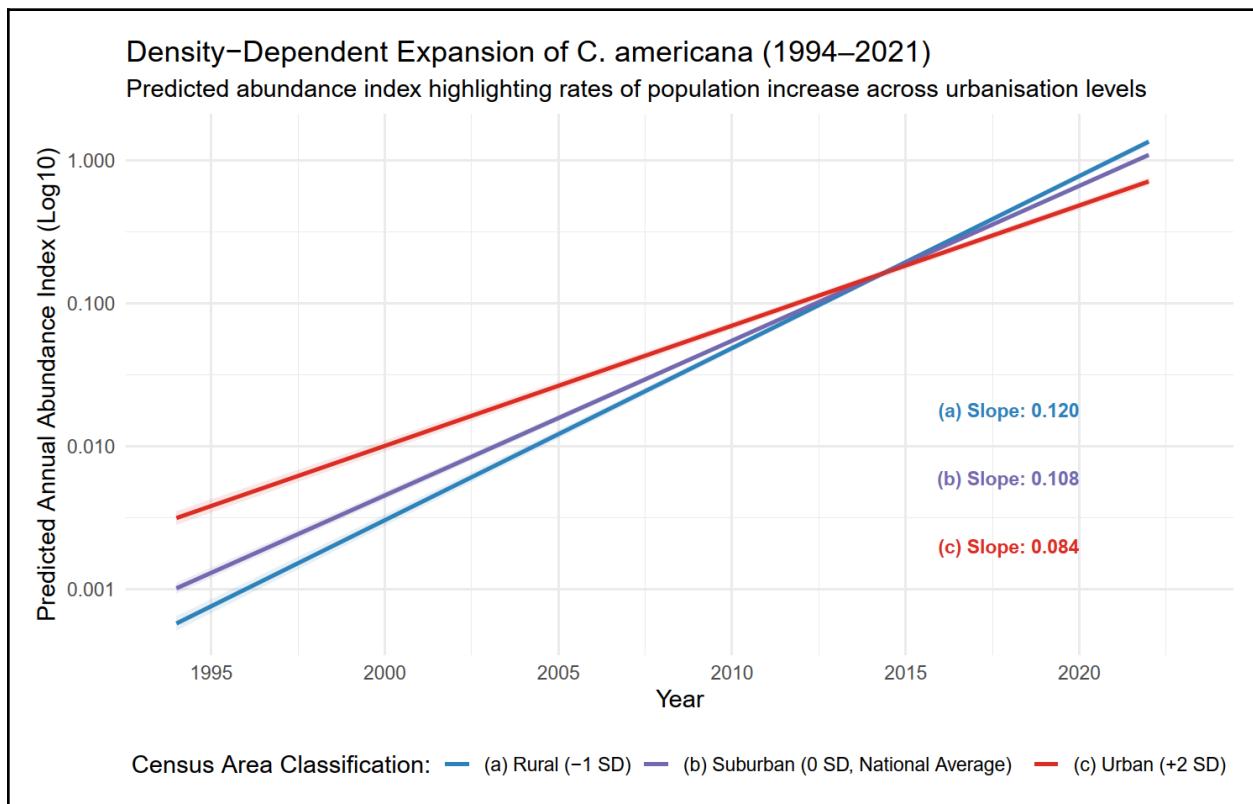


Figure 4: Predicted annual abundance index (PAAI) of *C. americana* between 1994 and 2021 across three simulated wards. Each represents a normalised human population density index ($Pop_Density_Z$) of: **(a)** -1 for a rural ward, **(b)** 0 (UK average) for a suburban one, and **(c)** 2 for an urban one.

The density-dependent phases of the invasion are illustrated through the interaction term ($Year_Centered \times Pop_Density_Z$), with all other variables controlled. The urban slope (fig. 4c) began with a higher PAAI than both the suburban (fig. 4b) and the rural one (fig. 4a). However, the latter slopes grow faster, denoted by their steeper logarithmic coefficients due to an interaction term's IRR inferior to 1. After converging in 2014, the rural slope exceeded the two others, finally positioning it with the highest PAAI, followed by the suburban one, then the urban one.

D. TEMPERATURE SENSITIVITY ANALYSIS (H4)

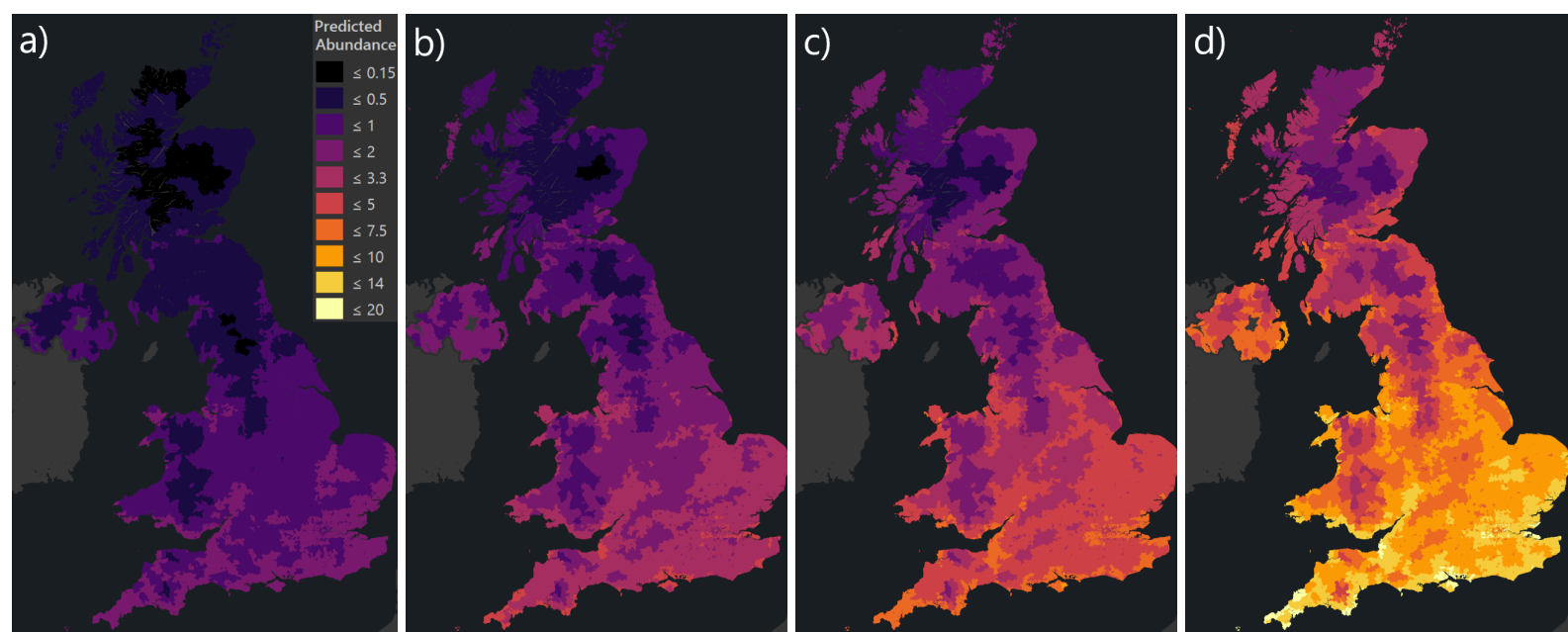


Figure 5: Spatial modeling of *C. americana* predicted annual abundance index (PAAI), which is its expected average sightings per year between 1994 and 2021 in each ward, under standardised recording effort. Projections under a uniform nationwide temperature increase above the 1981-2000 baseline are: **(a)** +0.0°C (mean = 0.767, median = 0.752, min = 0.057, max = 1.931, variance = 0.099) (baseline), **(b)** +1.5°C (mean = 1.776, median = 1.741, min = 0.132, max = 4.472, variance = 0.532), **(c)** +2.5°C (mean = 3.108, median = 3.048, min = 0.231, max = 7.829, variance = 1.632), **(d)** +4.0°C (mean = 7.199, median = 7.060, min = 0.536, max = 18.132, variance = 8.752).

The sensitivity analysis shows the isolated impact of temperature on *C. americana*'s abundance across climatic incremental scenarios (fig. 5b-d), based on a control scenario (fig. 5a). We observe the positive relationship between the PAAI and the national average temperature, which approximately doubles through each scenario. Even though the coldest regions experience more drastic shifts of temperature (due to its uniform application over the country), the warmest zones are the ones experiencing an explosion of the PAAI in the more advanced scenarios.

Southern England is the region that generally sees the highest PAAI, especially near zones where sea goes inland (Liverpool, Bristol Channel, Thames Estuary), and coastlines (with the Isle of Wight containing the warmest census area). On the other hand, inner Scotland scores the lowest index, followed by the Pennines region in Northern England, and inner Wales.

Finally, progressive warming is shown to increase the predicted abundance index in northern latitudes, while accentuating abundance inside cities, mostly Southern ones. Previously disconnected territories are now spatially merging (fig. 5d), linking the main areas of concentration, and forming more continuous corridors such as the border between England and Wales, or the English North-Eastern coastline.

IV. DISCUSSION

A. MAIN FINDINGS

A. 1. History of the invasion, between appearance and biological reality

A. 1. a. The dichotomy of population dynamics

C. americana's colonisation of the UK operates on two fronts, representing both the biological distribution of the species, and the spatial distribution of humans. There is a strong correlation between raw observations (fig. 2), and human population density (fig. 3b). Even historically, the earliest and largest zones to stand out on the charts are the urban centers like London, followed by Liverpool/Manchester. For a species that relies on conditions supported by the urban environment (UHI, and availability of host plants), human aggregation is a suitable, but flawed proxy metric. That proximity to human settlements conflates the biological extent of *C. americana* and its deemed abundance due to the REB, calling for the use of statistical tools to decouple these intertwining mechanisms.

Despite quantifying the same metric (number of observations), fig. 2 and 3a frame it distinctly, the former using general regional concentration, and the latter using wards. While Greater London as a region contains the most concentration of observations in the UK, their dilution among tiny wards minimises that representation. Overall, the average number of observations per ward is very low, at 2.863 (fig. 3a), most of them having one or none, making *C. americana* a relatively rare encounter across the UK, especially in Wales, Northern Ireland, and Scotland except for Edinburgh.

The rosemary beetle can also suffer from a detection lag phase, also known as the invasion debt, when performing jump dispersal to new areas (Rouget *et al.*, 2016): despite a successful establishment with a reproducing and sustainable population, the low likelihood of encounter (lack of recorders aware of its presence, local rarity...) may lead to a first census occurring years after its settlement in a region (Kelly *et al.*, 2021). A mismatch between perceived absence and biological absence could change the interpretation of invasion fronts to be actually further ahead than what datasets may suggest.

A. 1. b. Density coefficient as a metric

The density coefficient (DC) is a scale created for this study, measuring the proportion of observations in relation to local human population density. It aims to visualise the REB in action, at a ward, regional, and national scale. The DC helps identify areas where the beetle is exposed to recording biases in areas outside of regular settlements. Fig. 3c shows a DC that can be higher in zones that seem uninhabited, such as Cheshire and Yorkshire, which are localities situated near areas with high human density, while having low densities themselves. It could be due to the suburban nature of these regions, with a high density of gardens, or also because they are popular touristic, but non-urban areas to visit, leading to more observations (Geldmann *et al.*, 2016). East Anglia extends that notion, more wildlife and nature-oriented regions of the UK, where expeditions or targeted surveys happen can raise the number of observations outside the boundaries of densely populated cities (Boakes *et al.*, 2010).

In fig. 3c, the heterogeneity of DCs between wards in a region, such as South East England (located West and South of London), can be explained by the patchiness and variability of its component values in both fig. 3a and 3b, a mathematical artifact of their interactions: the DC, being a fraction, increases

alongside observations (numerator), but also when human density (denominator) decreases. The equal variation of both would output the same coefficient, as the formula is blind to absolute abundance, and only considers relative differences. This relationship is best exemplified on the coastal islands of Islay and Jura, in Western Scotland: despite their singular *C. americana* observation, their high DC is due to a minuscule human density.

Lastly, despite the numerous records in the capital city's general area (fig. 2), the overwhelming amount of inhabitants (and thus of observers) per ward in proportion to beetle observations creates a steep ratio, making populated hubs the lowest scoring regions after empty census areas. This supports H1, that more observations in cities are due to the concentration of inhabitants, inflating the records there. The interest in nature may also correlate negatively with more developed parts of the country (Bashan *et al.*, 2021), as despite more people partaking in recording there, they also represent a smaller proportion of the local population than they would in more suburban/rural places (Soga and Gaston, 2016).

A. 1. c. Introduction of a calibration species

The interactions between *C. americana* and *P. coccinea* can reflect the differences in their habitats, not only in resource suitability, but also in recorder enthusiasm, some preferring to scout the forested areas, while others the comfort of their own garden. The dissociation between human recording tendencies and urban habitat availability prevents their conflation. This polyvalence makes *P. coccinea* an adequate calibration species that isolates the REB from the rest of the predictors, both in the presence and abundance models.

In table 1, the baseline (average temperature, no *P. coccinea*) probability of detection of *C. americana* in any given ward is 9.8%, making it a relatively rare encounter on average. The odds of detection double, increasing that probability to 18.0%, when *P. coccinea* has also been detected in the same ward. This reflects human recording effort, active wards being more surveyed regardless of the species. However, table 2 inverts that relationship: for every additional observation of *P. coccinea*, the abundance of *C. americana* decreases by 17.1%, as they occur in different habitats. It is important to note that no direct ecological pressure is exerted onto the other (no predation nor competition for resources), effectively making both species covariates, subject to the same biases and predictors, without interfering with one another.

A. 2. The rosemary beetle across the urban-rural gradient

A. 2. a. Thermal constraints for settlement

The presence model (table 1) measured that an increase of 1 *SD* (0.859°C) over baseline in a ward makes *C. americana*'s likelihood of settling jump from 9.8% to 13.9%. This positive relationship associates the first occurrences of *C. americana* with the warmer areas of the country, more southern and/or under UHI influence. This supports H2, as temperature constitutes the primary constraint to the establishment of the species: host plants may be present in colder regions, but are not colonised due to the minimum thermal threshold required for survival over winter (Cox, 1994).

A. 2. b. Environmental drivers of abundance

Temperature has also been quantified as the second strongest metric, positively correlated with the count of *C. americana* in a ward at 61.8% for every 0.859°C (table 2). Likewise, optimal temperature conditions of cities enable the proliferation of a thermophilic species (Meineke *et al.*, 2013), as higher survival rates reinforce the exponential effect of population growth (Bale *et al.*, 2002). Human population density remains the strongest predictor of abundance, shifting by 76.1% for every 3,074 people/km²: being the habitat availability proxy, it generally increases with green space cover, and thus ornamental gardens offered by a ward (Gaston *et al.*, 2005).

Although human population density is also heavily associated with UHI, that effect is absorbed by the temperature predictor, effectively isolating both. The ambivalent effect of cities supports H3, anchoring the role of population density as the main driver of *C. americana* abundance thanks to the availability of habitat, followed by temperature, while controlling for recording effort bias.

A. 2. c. An approaching urban saturation

C. americana's abundance is also subject to time, reflecting a population that is still multiplying through reproduction, and colonisation of the territory through human-mediated jump dispersal, bringing new individuals into each ward over the years. This 28.3% annual rise at a national level (table 2) is uneven across the UK, due to the interaction term *Year_Centered* × *Pop_Density_Z*: a high *Pop_Density_Z* yields higher abundance at first, but the compounding effect of an *IRR* < 1 (0.973) disproportionately constrains areas with higher human population density over time, rendering its effect weaker annually.

Rates over time can be linked to the plant trade: resources for ornamental gardens were first in demand in major cities when *C. americana* first arrived (Dehnen-Schmutz *et al.*, 2007), serving as the primary and oldest nodes of the transport network (Hulme, 2009). This closely associates their initial population with anthropogenic activity, but over time, resource availability locally saturates, pressuring populations to seek other areas, similarly to studies done on other taxa (Matthysen, 2005). Simultaneously, demand for bushes in more remote cities and around the settled area increases over time, branching off the plant network, and thus the jump dispersal ability of the beetle, colonising smaller cities and their surroundings in turn (Suarez *et al.*, 2001).

Fig. 4 illustrates that density-dependence with the urban PAAI (fig. 4c) starting higher than the others, supporting the idea of cities as the cradle for *C. americana* upon its introduction in 1994. However, its rate of increase is more gentle than the others, due to the dampening effect of the interaction term's *IRR* on human population density, translating into a reduced rate of annual expansion, and effectively constraining urban beetle population growth as local resources reach saturation.

The plot suggests the transitional phases of population growth expanding outside of the dense urban areas as it shows signs of approaching carrying capacity, supporting H3. Pouring into the surrounding environments, the populations of beetles diffuse into suitable niches in the suburban areas (fig. 4b), rich in gardens (Loram *et al.*, 2007). A greater expansion happens in the rural habitats (fig. 4a), despite the rarer gardens, for two reasons: firstly, smaller initial populations are more sensitive to absolute increases, representing a higher relative percentage of growth rate than larger starting populations (Bale *et al.*, 2002). Secondly, the lack of intraspecific competition (Phillips, 2009), even when resources are not as abundant as the initial areas, makes the uncolonised territory a soaring ground for *C. americana* populations, resulting in a steeper logarithmic slope.

A. 3. Direction and forecast of the expansion

A. 3. a. A warming country

Saturation in cities mechanically phases out their monopoly on the beetle. The suburban and even rural areas are gradually becoming more viable, reinforced by temperature rises radiating away from the initial hot spots. In the spatial representations (fig. 5), we refocus the predicted annual abundance index (PAAI) to reflect changes of population in relation to the thermal predictor, forecasting potential abundance landscapes according to multiple climatic scenarios, effectively simulating the year 2021 if initial UK conditions were warmer.

In the baseline (fig. 5a), the southern coast is modelled to have the highest PAAI (1.931 observations per year between 1994 and 2021). Increasing temperature scenarios (fig. 5b-d) unlock habitat suitability towards the North, while accentuating abundance inside cities, mostly southern ones, supporting H4. Although the maps simulate the latest year of *C. americana* occurrences, the findings hint towards the progressive warming of a country that hosts a thermophilic species, which can be extrapolated into actual temperature increase models, mentioned further in the Significance section.

A. 3. b. Dispersal among corridors

The increasing PAAI over rural spaces (fig. 5) creates areas where *C. americana* can hold more stable populations, surviving and dispersing from in turn as long as suitable habitat can welcome them, supporting H4. The creation of these bridges and corridors, especially northwards and along the coastlines, is accentuated in the warmer scenarios, linking previously fragmented patches of higher abundance, a concurrent effect to the national average augmentation of the PAAI. And similarly to fig. 2g, islands of concentration that used to gravitate more central areas begin to aggregate, filling up the space between them (Shigesada *et al.*, 1995), a phenomenon also known as population coalescence (Liebhold and Tobin, 2008; Castledine *et al.*, 2020). Recursively, more islands where beetles can pool in are created, as a result of more city-hopping over time.

These formed corridors also improve the connectivity of the territory, by decreasing the distance required to travel between another patch of suitable habitat. Shorter distances facilitate a transition from a human-mediated long-distance dispersal, to a natural diffusion dispersal where the beetle can walk or even fly. The previously mentioned saturation in already colonised areas catalyses that process due to the pressure to find resources: diffusion requires a “comfort zone” where *C. americana* can still feed and reproduce, acting as a frontline of an invasion before it can successfully disperse beyond that sustainable boundary, also known as the bridgehead effect (Bertelsmeier and Keller, 2018). Areas that concentrate many ornamental gardens in proximity to each other act as stepping stones for the species’ local migration, by providing abundant intermediate resource pockets.

B. LIMITATIONS OF THE STUDY & POTENTIAL IMPROVEMENTS

B. 1. Citizen science & calibration

The *C. americana* monitoring scheme is bound by the constraints of any ecological survey. The recording effort bias, which we have discussed throughout, has been controlled to the best extent using the calibration species *P. coccinea*. Other biases intrude the recording process, notably the observer bias, which groups the individual perception of a recorder, often attributed to their identity or their means of recording (Warton *et al.*, 2013).

Before May 2008, the main data collectors were the members of the RHS, which accounted for 350,000 people (Royal Horticultural Society, 2012). This restriction spatially and behaviourally skewed the recorder base, limiting the observation pool during the exponential phase, a critical stage of the invasion where populations grow rapidly (Bale *et al.*, 2002). Members who have higher botanical/entomological expertise and knowledge are also more likely to be exposed to *C. americana* than the average person, potentially leading to over-reporting in localised areas.

The survey opened to the wider audience after 2008, coinciding with the launch of popular recording platforms like iNaturalist, boosting the population of recorders, and consequently inflating perceived abundance countrywide. REB may also express in recorder habitat preference, individuals more in line with forests, others with the convenience of their home, methodologically biasing records between model and target organisms. *C. americana* was also subject to a report campaign, accentuating efforts dedicated to finding the invasive species (known as the novelty bias (Goldstein and Stoudt, 2024)), whereas recordings of the native *P. coccinea* may be more opportunistic or passive without such urgency. The mixed effect of identity and methodological shift during the recording process of a dataset, also known as bipartition (Einav *et al.*, 2023), may lead to inconsistencies, and misrepresentation of data.

Moreover, zones of true biological absence affect presence-only observation data as a whole, conflating lack of recorders with lack of organisms, effectively creating blindspots. Although the impact on our target species is lesser due to its proximity to human infrastructure, this may be true for our calibration model *P. coccinea* living in remote areas. Species distribution models (SDMs) can bypass the limitation of unsurveyed areas, for example thanks to environmental similarity factors, but the most suitable model available for our opportunistic presence-only data, MaxEnt, does not predict abundance (Gomes *et al.*, 2018).

B. 2. Ecological accuracy of proxies & unexplained variance

We chose the habitat availability dataset to be approximated by human population density due to their positive interaction. Translated ecologically, the denser an area, the more beetles, regardless of the REB. However, this relationship may actually follow a bell curve distribution, where city centers lack garden infrastructure. More ideal proxies of habitat suitability can include green space maps, portraying the surface area of vegetation, narrowed down to ornamental/home gardens, more likely to correlate with the occurrence of Lamiaceae bushes. However, we did not find such specific dataset at the UK scale, only England, and while land use maps for the whole of the UK exist, they do not consider the specific garden habitat. Direct observations of the host plants exist too, but are way rarer, and more importantly are too fine to reflect on abundance on a ward scale, making them very inaccurate.

The temperature grid, despite its regular geometry, requires centroids to resolve conflicts between wards and squares overlapping. Shape/size differences also create a resolution mismatch: larger wards are coarser than the grid, stretching through multiple squares, and encompassing different possible temperatures. Meanwhile, smaller wards are finer, fitting inside a square, but will not consider the differences in microclimate caused by the UHI between wards from the same square, as it hosts different ranges of temperature across its 144 km².

The limitations caused by the use of proxies and the differences in resolutions are expressed in the *ICC* (table 2), showing that 23% of the residual variance in abundance was driven by inherent differences between the census areas that are not accounted for in the study: socio-ecologic factors like the surface area of green spaces, the number of Lamiaceae planted, the maintenance practices of gardens, microclimate, etc (Berthon *et al.*, 2021).

B. 3. Statistical isolation versus ecological realism

For the purposes of the study, human density and climate datasets were required to remain static: we disregarded the reality of fluctuating demography or temperature across time, and their temporal mismatch. Dynamic models, on top of being unsuitable due to the climate tracking lag of the beetle (the population's adaptation time to new conditions) (Chen *et al.*, 2023), would interfere with the statistical process: the mathematical assessment of single predictor impacts is an essential component of statistical studies (Pourhoseingholi *et al.*, 2012). Besides, the seeming lack of interaction can be reinjected in the study using the formula associated with the models from their respective Methods section.

However, data represented to underline a particular aspect can indeed make that lack of interaction resurface: the focalisation on select variables may sacrifice to an extent the understanding of the general picture, if the holistic communication of information is the main purpose. For example, in fig. 4, the plot displays an overtake of the rural curve in 2014: at first glance, the rural areas seem to have more abundance by the end. But the nullification of the other variables, while statistically sound, misrepresents the thermal advantage of cities, which would have shifted the urban slope vertically upwards if it were not stripped away, and the lines would never cross.

Similarly in fig. 5, one variable was isolated and modulated, simulating adjusted climates and how they impacted the abundance index. While the raw dataset contained pre-calculated values for the incremental temperature changes, they represented international increases, which would not necessarily affect each ward equally due to dynamically calculated values (taking into account different amounts of warming based on urbanisation, latitude, altitude...). Thus, we opted for a flat, uniform increase from a baseline to properly isolate the temperature predictor.

C. SIGNIFICANCE & PROSPECTS

C. 1. Methodological significance

The integration of larger datasets into scientific analyses was enabled by the participation of contributors throughout the country, facilitating the accrual of biological observations at a national scale. When methodologically standardised, this bypasses the limitations of sampling size and geographical extent. The consequent recording and observer biases are pervasive to such studies, and would highly influence results to represent human presence rather than the model organism being investigated. This highlights the use of techniques to mitigate them, like the integration of a calibration species, useful through their conspicuity and the analysis of their co-occurrence or not with the target organism. A framework integrating that aspect, coupled with NB-GLMMs accounting for mechanisms like overdispersion and pseudoreplication, creates a replicable pipeline for similar research, transforming citizen science datasets into interpretable abundance models that reflect biological truth rather than human influence.

C. 2. Ecological significance

The study, as previously mentioned, can serve as a model for understanding dispersion of invasive species, notably thermophilic, jump-dispersing pests, or just ones associated with the rural-urban gradient. The study highlights the evolving role of urban infrastructure on population dynamics, from the incubation phase, its development, to a full-scale invasion branching into other regions. It also reflects on the use of habitats and landscapes, exposing implications of the effect of urban expansion onto such populations, and the shift in climate associated with it through the UHI. The simulated thermal models can be reattached to forecasted climate scenarios, like the Shared Socioeconomic Pathways (SSPs) proposed by the Intergovernmental Panel on Climate Change (IPCC), as the temperature sensitivity analysis can be extrapolated into temperature models associated with a specific decade, used in determining global trends to better develop environmental programs.

C. 3. Accessibility of the research to wider audiences

The final prospect of our study, in accordance with our aims, is to reach out to audiences through more modern means. The communication behind invasion ecology may be buried into mathematical and statistical modelling, an approach not always tailored for a more layman audience. Even a public with basic notions in biology requires more accessible exposure to scientific findings, and domains rooted in biogeography benefit from visual communication of the areas impacted, to better represent areas of sensitivity deserving of preoccupation. Interactive and user-friendly online platforms can host the contents of a research paper alongside narrative-driven cartographic visualisation, bridging the gap between scientific concepts and their real-life application.

The direct beneficiaries of this research are people impacted by the beetle's spread, mainly regular gardeners (around 42% of the British population (Bisgrove and Hadley, 2002)), and citizen scientists seeking a deeper understanding of the species distribution. More broadly, research on invasion ecology as a whole benefits political entities (like councils), NGOs, and stakeholders in policy making. Conveying insights, statistics, and the science behind the interactions between organisms and (a)biotic factors can be a challenge when the organisations in charge balance funding versus the ecological benefits of habitat management, so the streamlining of this communication facilitates the decision-making process.

V. CONCLUSION

Observer effort is a major component of citizen science datasets, that can misrepresent findings if not accounted for. We demonstrated that species tethered to human presence due to habitat suitability (temperature and food source) are even more prone to that bias, requiring the intervention of statistical tools to decouple their independent effect (H1). The rosemary beetle, which arrived in the UK in the 1990s, found suitable habitat in the cities that provided the necessary thermal refuges for its establishment (H2), reminiscent of the conditions of its home territory in the Mediterranean region.

The thermal advantage of cities was also linked to the higher abundance of *C. americana*, coupled with the fact they were the first areas to be colonised due to the prominence of their host plants, members of the family Lamiaceae that move from urban hubs to another because of the plant trade. Although cities were the primary grounds for the initial expansion of *C. americana* populations, we mathematically uncovered the existence of a carrying capacity that exhausted that growth rate over the course of several years. This drove a more exponential growth into surrounding suburban and rural spaces, where the beetle found more abundant resources, and the lack of intraspecific competition (H3).

Our last analysis tested for the sensitivity of the beetle to national temperature increase, to display a predicted abundance simulated under different initial climatic conditions (H4). This resulted in more predicted abundance in cities, and in the increasing suitability of rural habitats that homogenise the distribution of the beetle throughout the country, facilitating expansion beyond the fragmented islands formed by the urban heat island effect.

Ultimately, the rosemary beetle serves as an excellent model organism to study the invasion ecology and population dynamics of a species that only appeared 30 years ago in the UK. Records of its presence since its introduction are useful to quantify and map its occurrence, even when blurred by biases accompanying citizen science, in which case statistical models help in accounting for such limitations.

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SUPPLEMENTARY INFORMATION

A. RAW DATASETS

A. 1. Beetle observations

Source: RHS on GBIF for *C. americana* ([link](#)), and various on GBIF for *P. coccinea* ([link](#)).

Cleaned tables of *C. americana* ("Chryso"), and of *P. coccinea* ("Pyro") variables:

- **Chryso/Pyro_id**: unique identifier of the beetle observation.
- **Chryso/Pyro_lat**: latitude of the observation, according to the WGS 1984 Web Mercator projected coordinate system.
- **Chryso/Pyro_lon**: longitude of the observation.
- **Chryso/Pyro_year**: year of the observation.

Chryso_id	Chryso_lat	Chryso_lon	Chryso_year
1	51.29652	-0.38782	2000
2	51.49673	-0.15242	2001
3	51.45876	-0.19714	2001
4	51.57432	-0.05692	2001

Pyro_id	Pyro_lat	Pyro_lon	Pyro_year
1	51.55745	-3.22748	1897
2	50.84399	-1.64629	1909
3	50.84399	-1.64629	1919
4	52.07649	-2.96078	1936

A. 2. Temperature grid

Source: Met Office ([link](#)).

Cleaned table of variables:

- **square_id**: unique identifier of the 12×12 km grid square.
- **square_projection_y_coordinate**: latitude of the square, according to the British National Grid projected coordinate system. The table also contained another latitude column, which was relating to another coordinate system (the WGS 1984 Web Mercator, expressed in degrees), discussed further down in the GIS Workflow section.
- **square_projection_x_coordinate**: longitude of the square.
- **baseline_1981_2000**: annual average temperature recorded between 1981 and 2000 in that particular square.

square_id	square_projection_y_coordinate	square_projection_x_coordinate	baseline_1981_2000
1	18000	138000	10.74
2	18000	150000	11.14
3	18000	162000	10.97
4	18000	174000	10.81

A. 3. Census areas

Source: ONS for England/Wales ([link](#)), Scottish Government for Scotland ([link](#)), NISRA for Northern Ireland ([link](#)).

Cleaned tables of England/Wales, Scotland, and Northern Ireland variables:

- **Zone_Code/IZCode/SDZ2021_cd**: unique code identifier of the census area.
- **Zone_Name/IZName/SDZ2021_nm**: name of the census area.
- **Total_Pop/TotPop2022/All_usual_residents**: human population of the area.
- **Area_SqKm/StdAreaKM2**: surface area of the census area in km².
- **Pop_Density_SqKm/Population_density_km²**: population density of the census area.

Zone_Code	Zone_Name	Total_Pop	Area_SqKm	Pop_Density_SqKm
E05000932	Ainsdale	12676	8.410194	1507.218
E05000933	Birkdale	12700	3.064401	4144.366
E05000934	Blundellsands	11464	4.889613	2344.562
E05000935	Cambridge	12466	9.842556	1266.541

IZCode	IZName	TotPop2022	StdAreaKM2	Population_density_km ²
S02002515	Culter	4753	19.71563	241.0778
S02002516	Cults, Bieldside and Milltimber West	4864	16.99166	286.2581
S02002517	Cults, Bieldside and Milltimber East	3465	3.933389	880.9197
S02002518	Cults, Bieldside and Milltimber Central	3388	3.523136	961.6433

SDZ2021_cd	SDZ2021_nm	All_usual_residents	Area_SqKm	Population_density_km ²
N21000001	Dunsilly_A	3272	66.1141	49.49024
N21000002	Dunsilly_B	3132	72.9619	42.92652
N21000003	Dunsilly_C	2527	27.9437	90.43169
N21000004	Dunsilly_D	2991	1.1161	2679.77

B. DETAILS OF ANALYTICAL PROCESSES

B. 1. Coding script pipeline

The entire R script pipeline used for the project is provided externally (R and pdf format), through GitHub.

([link to GitHub](https://github.com/K0rosan/MSc-TBE-Thesis---Rosemary-Beetle), <https://github.com/K0rosan/MSc-TBE-Thesis---Rosemary-Beetle>)

B. 2. Elaboration of methods executed in R

Preliminaries for statistical models	The package <i>stringr</i> was used with the command: • <code>str_replace_all()</code> to remove empty spaces that could contaminate text data. The package <i>dplyr</i> was used mainly with the commands: • <code>left_join()</code> to unify the geographic and demographic data, using the communal zone code in their respective table. • To convert the geometric data like <code>Shape_Area</code> to turn m^2 into km^2, which was then processed in section II. B. 2. • <code>filter()</code> (with argument <code>!is.na()</code>) to remove polygons that lack a human population from the dataset, resulting in 9,792 areas. Also used for the statistical models in H2 and H3, to filter out wards that do not contain either an observation of <i>C. americana</i> nor <i>P. coccinea</i>, resulting in the consideration of 5,325 wards. The package <i>sf</i> was used mainly with the commands: • <code>st_transform()</code> to standardise the coordinate system projection, from the WGS 1984 Web Mercator (auxiliary sphere) (EPSG: 3857), to the British National Grid (EPSG: 27700) across every dataset. • <code>st_centroid()</code> to bind the geographic centroid of wards onto the corresponding Met Office grid squares. • <code>st_drop_geometry()</code> to bypass unrequired geometric calculation for processing speed. • <code>st_join()</code> to capture beetle observations within the boundaries of each ward using a spatial join.
Presence model	The base command <code>glm()</code> executes the binomial generalised linear model (B-GLM): <pre> thermal_model <- base::glm(Presence_Ca ~ # Target species detection (1 or 0) Temp_Hist_Z + # Historical mean temperature Presence_Pc, # Calibration species detection (1 or 0) family = binomial(link = "logit"), # Logistic regression data = master_panel) </pre>

	The binomial family of GLMs extends standard linear regression by allowing for non-normal error distributions, calculating a two-state (binary) proportion data using binomial distribution, instead of a continuous data using Gaussian distribution. The model is coupled with a logit link function to transform probability values into continuous log-odds. Because the coefficients are calculated on a logarithmic scale, the package <i>sjPlot</i> was used with the command <i>tab_model()</i> to generate a summary table that automatically exponentiates these coefficients (formula e^{β}) to obtain the odds ratio. The <i>OR</i> expresses how much the baseline odds of a binary response variable are multiplied by when a given independent variable (predictor or covariate) increases by one unit (of count or of standard deviation for Z-scores). An <i>OR</i> > 1 multiplies these odds positively, whereas multiplication with an <i>OR</i> < 1 decreases those odds. In context, the <i>OR</i> expresses how much the odds of detecting <i>C. americana</i> (binary state, presence or absence) in a ward are multiplied by for every standard deviation of temperature, and also by how much these odds differ if <i>P. coccinea</i> is also detected (also presence or absence).
Abundance model	The package <i>glmmTMB</i> was used with the command <i>glmmTMB()</i> to execute the negative binomial generalised linear mixed model (NB-GLMM): <pre> abundance_model_tmb <- glmmTMB::glmmTMB(Count_Ca ~ # Target species abundance Year_Centered * # Time elapsed (Year 0 = 1994) Pop_Density_Z + # Human population density Temp_Hist_Z + # Historical mean temperature Count_Pc + # Calibration species abundance (1 Zone_Code), # Random spatial intercept family = nbinom2, # Negative binomial distribution data = master_panel) </pre> Negative binomial GLMMs are statistical models accounting for overdispersion of data. The model uses the second parametrisation family of negative binomials (<i>nbinom2</i>), which accounts for a variance (σ^2) that increases quadratically with the mean (μ). This describes the nature of population clustering better than the first family, <i>nbinom1</i>, for when variance increases linearly with the mean. The model is inherently coupled with a log link function, to ensure count data remains positive, and to transform the additive effects of the predictors on the logarithmic scale into multiplicative effects on the response scale. Similarly to the presence model, because the coefficients are calculated on a logarithmic scale, the package <i>sjplot</i> exponentiated these coefficients to obtain the incidence rate ratios (<i>IRR</i>).

	The <i>IRR</i> expresses the multiplicative change of a response variable exerted by a one unit (of count or of standard deviation for Z-scores) variation of an independent variable (predictor or covariate). An <i>IRR</i> > 1 multiplies the response variable positively, whereas multiplication with an <i>IRR</i> < 1 decreases it. In context, the <i>IRR</i> expresses the sensitivity of <i>C. americana</i> abundance to every additional/fewer standard deviation of temperature and of population density, while also reacting to every extra increase in year, and occurrence of <i>P. coccinea</i>. The <i>IRR</i> for the interaction term (effect of population density over time) is also described, to expand onto the relationships between predictors, not only direct influences towards the response variable.
Interaction model	The package <i>ggeffects</i> was used with the command <i>ggpredict()</i> to extract the interaction term, and projecting the PAAI over time: <pre> predictions_tmb <- ggeffects::ggpredict(abundance_model_tmb, terms = c("Year_Centered", # Plotting vs time on the x-axis "Pop_Density_Z [-1, 0, 2]" # Assigning the urbanisation values), condition = c(Count_Pc = 0) # Assuming no observer bias) </pre> The argument <i>Year_Centered</i> is placed first, to define it as the x-axis, while the argument <i>Pop_Density_Z</i> is second to represent the modulated variable: it has been assigned arbitrary urbanisation levels, that do not necessarily correspond to our real dataset range (<i>Pop_Density_Z</i> min = -0.854, max = 8.337), but just the general directions of trends. In the <i>ggpredict()</i>, variables from the inserted model (<i>abundance_model_tmb</i>) that are omitted are averaged, which means they do not get controlled for by default. This is why we use the condition argument to simulate a baseline abundance of <i>P. coccinea</i>, but not for <i>Temp_Hist_Z</i> (as its average is already 0, thus already controlled for).
Climate model	The base command <i>predict()</i> executes the baseline climate simulation (increase of +0°C), acting as the foundation for the alternate climate scenarios: <pre> future_predictions_table\$Pred_Abund_Baseline <- base::predict(abundance_model_tmb, # feeding the abundance model newdata = base_pred_data, # feeding the temperature values data type = "response", # output is a count number re.form = NA # marginal prediction) </pre>

	Then the next script loops for every value of increment (<i>temp_inc</i> = 1.5, 2.5, then 4.0): <pre>temp_scenario_data <- base_pred_data %>% dplyr::mutate(# formula described in section II. E Temp_Hist_Z = ((Temp_Hist + temp_inc) - temp_mean) / temp_sd) # Then rerun the predict() command, replacing base_pred_data with temp_scenario_data</pre> The marginal prediction (argument <i>re.form</i> = <i>NA</i>) strips away the random effects (<i>Zone_Code</i>), or the ward-specific variances (replicating the conditions for an average baseline ward) for the climate simulations. The package <i>sf</i> was used with the command <i>st_write()</i> to export table/geometric data associated with each ward into a GeoPackage, for the cartographic rendering of the results.
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B. 3. Mathematical contextualisation of the Z-scores

Standard deviation	A standard deviation (<i>SD</i>) is a statistical unit used to quantify the amount of variation in a set of values relative to their mean. The model converts variance (noted σ^2, which is the average of the squared differences from the mean) into standard deviation by taking its square root (σ, or <i>SD</i>). The <i>SD</i> for the relevant ward values (temperature and population density) are then converted into Z-scores of our study (<i>Temp_Hist_Z</i> and <i>Pop_Density_Z</i>) to quantify these predictors on a normalised scale. The statistical model assigns a null score that corresponds to the datasets average, and not an absolute measure of 0 in the predictor's unit. For example, a ward scoring 0 in <i>Temp_Hist_Z</i> signifies it is average in terms of temperature when considering the whole country, and not 0°C. Likewise, a null value in <i>Pop_Density_Z</i> means an average human population density, and not the absence of people. This also allows a more fair statistical comparison between the predictors, as the <i>SD</i> considers distance from the average as the unit. Our models can also reconvert the <i>SD</i> into their corresponding unit (°C and people/km²), translating the statistical units of our study into real-term equivalents. For example, 1 <i>SD</i> of temperature has been calculated to be 0.859°C away from the national average (in both presence and abundance models). Thus, -1 <i>SD</i> = 0.859°C below national average, 2 <i>SD</i> = 0.859 × 2 = 1.718°C above average, etc. The compounding effect of standard deviation, just like any unit of independent variable, highlights the exponential, rather than linear, effect of predictors.
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Presence model	It is to be noted that because the measure of presence is binary (thus the choice of a binomial model), it is clearer to represent these outputs over a large number of wards. Odds and probabilities (obtained through the <i>logit</i> function of the <i>glm()</i> command) give a tangible representation of the likelihood of establishment of <i>C. americana</i> in a ward. For 1,000 simulated wards of average temperature and no <i>P. coccinea</i> observation, the odds of establishment are 0.109 to 1, which is a probability of $0.109 / (1 + 0.109) = 0.098$. Thus, 98 wards on average (9.8% of 1,000) will contain at least one <i>C. americana</i> observation. Variation in standard deviation affect the odds in the following ways: <table><tr><td><i>SD</i></td><td>-1</td><td>0</td><td>+1</td><td>+2</td></tr><tr><td>Measurement</td><td>-0.859°C</td><td>±0°C</td><td>+0.859°C</td><td>+1.718°C</td></tr><tr><td>Odds formula</td><td>0.109×1.487^{-1}</td><td>0.109×1.487^0</td><td>0.109×1.487^1</td><td>0.109×1.487^2</td></tr><tr><td>Odds</td><td>0.073</td><td>0.109</td><td>0.162</td><td>0.241</td></tr><tr><td>Prob. formula</td><td>$0.073 / (1 + 0.073)$</td><td>$0.109 / (1 + 0.109)$</td><td>$0.162 / (1 + 0.162)$</td><td>$0.241 / (1 + 0.241)$</td></tr><tr><td>Probability</td><td>0.068</td><td>0.098</td><td>0.139</td><td>0.194</td></tr><tr><td>Mean presence (for 1,000 wards)</td><td>68</td><td>98</td><td>139</td><td>194</td></tr></table>	<i>SD</i>	-1	0	+1	+2	Measurement	-0.859°C	±0°C	+0.859°C	+1.718°C	Odds formula	0.109×1.487^{-1}	0.109×1.487^0	0.109×1.487^1	0.109×1.487^2	Odds	0.073	0.109	0.162	0.241	Prob. formula	$0.073 / (1 + 0.073)$	$0.109 / (1 + 0.109)$	$0.162 / (1 + 0.162)$	$0.241 / (1 + 0.241)$	Probability	0.068	0.098	0.139	0.194	Mean presence (for 1,000 wards)	68	98	139	194
<i>SD</i>	-1	0	+1	+2																																
Measurement	-0.859°C	±0°C	+0.859°C	+1.718°C																																
Odds formula	0.109×1.487^{-1}	0.109×1.487^0	0.109×1.487^1	0.109×1.487^2																																
Odds	0.073	0.109	0.162	0.241																																
Prob. formula	$0.073 / (1 + 0.073)$	$0.109 / (1 + 0.109)$	$0.162 / (1 + 0.162)$	$0.241 / (1 + 0.241)$																																
Probability	0.068	0.098	0.139	0.194																																
Mean presence (for 1,000 wards)	68	98	139	194																																
Abundance model	To illustrate <i>Temp_Hist_Z</i>, modulating temperature for an otherwise average ward with null predictors that contains 1000 observations of rosemary beetles, with an <i>IRR</i> = 1.618: <table><tr><td><i>SD</i></td><td>-1</td><td>0</td><td>+1</td><td>+2</td></tr><tr><td>Measurement</td><td>-0.859°C</td><td>±0°C</td><td>+0.859°C</td><td>+1.718°C</td></tr><tr><td>Formula</td><td>1000×1.618^{-1}</td><td>1000×1.618^0</td><td>1000×1.618^1</td><td>1000×1.618^2</td></tr><tr><td>Abundance</td><td>618</td><td>1000</td><td>1618</td><td>2618</td></tr></table> Note: this value of <i>IRR</i> (1.618) corresponds to the golden ratio, which means its exponentiation using the <i>SD</i> will result in very regular sequences that make the relationships seem linear (618, 1618, 2618...), which is a property of that number. This has been confirmed not to be a mathematical error of our study, as they are slightly different (golden ratio $\phi \approx 1.618033$, while our <i>IRR</i> ≈ 1.617530). It is thus a coincidence, as other datasets with different parameters would output other <i>Temp_Hist_Z IRR</i> values, not necessarily rooted into an ecological explanation, although the concept can be found in other areas of population ecology (like growth rates and the Fibonacci sequence). Similarly for <i>Pop_Density_Z</i>, modulating human population density for an otherwise average ward with null predictors that contains 1000 observations, with an <i>IRR</i> = 1.761:	<i>SD</i>	-1	0	+1	+2	Measurement	-0.859°C	±0°C	+0.859°C	+1.718°C	Formula	1000×1.618^{-1}	1000×1.618^0	1000×1.618^1	1000×1.618^2	Abundance	618	1000	1618	2618															
<i>SD</i>	-1	0	+1	+2																																
Measurement	-0.859°C	±0°C	+0.859°C	+1.718°C																																
Formula	1000×1.618^{-1}	1000×1.618^0	1000×1.618^1	1000×1.618^2																																
Abundance	618	1000	1618	2618																																

	<table><tr><td><i>SD</i></td><td>-1</td><td>0</td><td>+1</td><td>+2</td></tr><tr><td>Measurement</td><td>-3,074 ppl/km²</td><td>±0 ppl/km²</td><td>+3,074 ppl/km²</td><td>+6,148 ppl/km²</td></tr><tr><td>Formula</td><td>1000 × 1.761⁻¹</td><td>1000 × 1.761⁰</td><td>1000 × 1.761¹</td><td>1000 × 1.761²</td></tr><tr><td>Abundance</td><td>568</td><td>1000</td><td>1761</td><td>3101</td></tr></table>	<i>SD</i>	-1	0	+1	+2	Measurement	-3,074 ppl/km²	±0 ppl/km²	+3,074 ppl/km²	+6,148 ppl/km²	Formula	1000 × 1.761 ⁻¹	1000 × 1.761 ⁰	1000 × 1.761 ¹	1000 × 1.761 ²	Abundance	568	1000	1761	3101
<i>SD</i>	-1	0	+1	+2																	
Measurement	-3,074 ppl/km²	±0 ppl/km²	+3,074 ppl/km²	+6,148 ppl/km²																	
Formula	1000 × 1.761 ⁻¹	1000 × 1.761 ⁰	1000 × 1.761 ¹	1000 × 1.761 ²																	
Abundance	568	1000	1761	3101																	
Interaction model	The effect of population density is ambivalent: it affects both the response variable (abundance of <i>C. americana</i>) and another predictor (time), which means the annual rate of expansion (but not the <i>IRR</i>) of <i>Year_Centered</i> actually strays away from 1.283 depending on the value of <i>Pop_Density_Z</i>. The interaction term (<i>Year_Centered</i> × <i>Pop_Density_Z</i>) being a multiplication, is symmetrical, meaning processes that can be applied to one can also be applied to the other, depending on the aims for the analysis. For example, ours plotted the effect of <i>Pop_Density_Z</i> over time, so the <i>IRR</i> of the <i>Year_Centered</i> variable was the one being isolated in front, as to give the formula 1.283 × 0.973^{<i>Pop_Density_Z</i>}. Which means an alternate study could have isolated urbanisation level, with the value of the year being modulated (formula 1.761 × 0.973^{<i>Year_Centered</i>}). The result would have been a plot with the x-axis being the values of <i>Pop_Density_Z</i>, and each curve representing a different year. Some examples of <i>Pop_Density_Z</i> values effect on the annual rate of expansion: <table><tr><td><i>SD</i></td><td>-1 (rural)</td><td>0 (suburban)</td><td>+1</td><td>+2 (urban)</td></tr><tr><td>Formula</td><td>1.283 × 0.973⁻¹</td><td>1.283 × 0.973⁰</td><td>1.283 × 0.973¹</td><td>1.283 × 0.973²</td></tr><tr><td>Annual rate</td><td>1.319</td><td>1.283</td><td>1.248</td><td>1.215</td></tr></table> Negative values of <i>SD</i> for <i>Pop_Density_Z</i> make the annual rate of expansion actually increase, while positive ones lower it, therefore constraining to a greater extent the urban curve due to its association with human population density.	<i>SD</i>	-1 (rural)	0 (suburban)	+1	+2 (urban)	Formula	1.283 × 0.973 ⁻¹	1.283 × 0.973 ⁰	1.283 × 0.973 ¹	1.283 × 0.973 ²	Annual rate	1.319	1.283	1.248	1.215					
<i>SD</i>	-1 (rural)	0 (suburban)	+1	+2 (urban)																	
Formula	1.283 × 0.973 ⁻¹	1.283 × 0.973 ⁰	1.283 × 0.973 ¹	1.283 × 0.973 ²																	
Annual rate	1.319	1.283	1.248	1.215																	
Climate model	The standard deviations of the temperature values in the baseline are different from the ones in the incremental scenarios (thus a 2.5°C temperature increase would not correspond to an increase of roughly 3 <i>SD</i> (2.5°C / 0.859°C), etc). When creating the incremental scenarios, the Z-score needs to adapt to match the new range, mean and variance. However, stopping there and skipping to the spatial representation would output four identical maps, as there is no relative difference between the wards (and the national average would simply reset to 0). To fix that, we need to reuse the original mean and variance, anchoring the baseline as the reference for change, to properly treat the scenarios as anomalies (formula described in II. E).																				

C. GIS WORKFLOW

C. 1. Datasets visualisation & coordinate projections

All datasets were standardised to the British national grid (EPSG: 27700) during the analyses, to preserve the size of the census area, as well as the perpendicularity of the temperature grid. In ArcGIS Pro, the tool “Table to Point” was used on the *C. americana*, *P. coccinea*, and Temperature Grid datasets to convert their coordinates into presentable spatial points, before applying their unique symbology.

Temperature Grid was taken further by applying the tool “Create Thiessen Polygon” to convert the cardinally arranged points into squares, creating a 12x12km grid covering the entire UK. A byproduct of this process is the capture of surrounding areas as it draws a rectangle around the perimeter of interest. However, this includes the sea, which we removed with the “Clip” tool, using the country’s borders as the cookie-cutter (layer “GBR_Country”, imported from ArcGIS’ collection of geographic information, the Living Atlas of the World), to only consider inland.

C. 2. Results visualisation (density & climate scenario maps)

The R code outputs a Geopackage containing the geometries of all census areas (displayed on a map layer), as well as their attributes (displayed in a table, each with their own column). In ArcGIS Pro, individual layers based on the relevant attributes were then created, to which their respective symbology was assigned (section II. B. 2. and II. E.). The attribute table can then be exported into this csv:

Zone_Code	Zone_Name	Total_Pop	Area_Sq_Km	Pop_Density_SqKm	Pop_Density_Z	Total_Ca	Density_Coefficient	Pred_Aband_Baseline	Pred_Aband_15C	Pred_Aband_25C	Pred_Aband_40C	Dummy_Scale
E05000932	Ainsdale	12676	8.410194	1507.218	-0.36425	20	0.013269	0.89059	2.062642	3.610672	8.362463	0
E05000933	Birkdale	12700	3.064401	4144.366	0.493563	8	0.00193	0.759541	1.759129	3.07937	7.131946	0.002043
E05000934	Blundellsands	11464	4.889613	2344.562	-0.09188	20	0.00853	0.905542	2.097272	3.671291	8.502861	0.004085
E05000935	Cambridge	12466	9.842556	1266.541	-0.44253	10	0.007896	0.903621	2.092824	3.663505	8.484827	0.006128

The attribute table’s fields list, with the data information and description:

Column name	Data type & (domain interval)	Description
Zone_Code	categorical nominal (N/A)	Unique code identifier of the census area, starting with its country code.
Zone_Name	categorical nominal (N/A)	Name of the census area.

Total_Pop	numerical discrete ([0; +∞])	Estimated count of human population of the census area.
Area_SqKm	numerical continuous ([0; +∞])	Surface area of the census area in km ² .
Pop_Density_SqKm	numerical continuous ([0; +∞])	Population density of the census area, calculated using <i>Total_Pop</i> / <i>Area_SqKm</i> . Used to generate fig. 3b.
Pop_Density_Z	numerical continuous ([-∞; +∞])	Population density as a normalised Z-score (value of 0 represents the country's average). Also could be used to generate fig. 3b, as the relative differences between wards are the same in both columns.
Total_Ca	numerical discrete ([0; +∞])	Observed count of <i>C. americana</i> in the census area. Used to generate fig. 3a.
Density_Coefficient	numerical continuous ([0; +∞])	Density coefficient (DC) of the census area, calculated using <i>Total_Ca</i> / <i>Pop_Density_SqKm</i> . Used to generate fig. 3c.
Pred_Abund_Baseline	numerical continuous ([0; +∞])	Predicted annual abundance index (PAAI) of <i>C. americana</i> observations for the year 2021, driven by historical regional climate (1981–2000 averages), while accounting for REB. Used to generate fig. 5a.
Pred_Abund_15C	numerical continuous ([0; +∞])	PAAI of <i>C. americana</i> observations for the year 2021, assuming a uniform +1.5°C atmospheric warming above the historical baseline across the wards. Used to generate fig. 5b.
Pred_Abund_25C	numerical continuous ([0; +∞])	Ditto, but for a uniform warming increase of +2.5°C above the baseline. Used to generate fig. 5c.
Pred_Abund_40C	numerical continuous ([0; +∞])	Ditto, but for a uniform warming increase of +4.0°C above the baseline. Used to generate fig. 5d.
Dummy_Scale	numerical continuous ([0; +∞])	Assigned value between 0 and 20 to encompass every calculated value of PAAI (0.536 to 18.132). This allows harmonisation, and fair comparison between the symbologies of the four climatic simulations.

C. 3. High resolution mapping: ArcGIS Online/StoryMaps

The visualisation of cartographic data is more appreciated in higher resolutions. This is why we included a couple of options to access vectorial spatial representation, which allows zooming without sacrificing the quality of the image, as well as adaptive boundary features (dynamic line widths).

Exporting spatial representations to ArcGIS Online (with the feature “Share As Web Layer”) switches the projected coordinate system to WGS 1984 Web Mercator (auxiliary sphere) (EPSG: 3857) to match the online hosted platform. Exporting maps from the desktop application ArcGIS Pro to the online hosted application ArcGIS online allows the use of our spatial data across the Esri environment. That includes ArcGIS StoryMaps, a platform designed for presentations centered around the communication of geographic data.

These extra materials are accessible through the GitHub repository accompanying this project:

([link to GitHub](https://github.com/K0rosan/MSc-TBE-Thesis---Rosemary-Beetle), <https://github.com/K0rosan/MSc-TBE-Thesis---Rosemary-Beetle>)